BINARIOS

SUMA

1- 1110
1101
11011

2- 1101
1000
10101

3- 1000
1111
10111

4- 1101
11
10000

5- 11110000
1111000
111100000

6- 10001111
1111
10011110

7- 10000000
11111111
101111111

8- 10000011
11110001
101110100

9- 10101011
11111111
110101010

10- 10010010
10101010
100111100

RESTA

a- 1110
1101
0001

b- 1101
10
1011

c- 1000
11
0111

d- 1001
101
0100

e- 100
11
001

f- 11111001
10001111
01101010

g- 11100111
11000
11001111

h- 10101010
1010101
01010101

i- 10111100
1010111
01100101

j- 11100011
1000111
10011100

MULTIPLICACIÓN

1- 11110101
110
111101010
11110101
10110111110

2- 10000111
101
10000111
10000111
1010100011

3- 10000011
11100011
10000011
10000011
1000001
100000
100000
11100110101001

4- 10101010
1010101
10101010
10101010
10101010
10101010
11101011110010

5- 10000111
10001111
10000111
10000111
10000111
10000111
10000111
100101110111101

6- 10101100
10001
10101100
10101100
101101101100

7- 11100011
10011111
11100011
11100011
11100011
11100011
11100011
11100011
1011111111111101

8- 10101001
10000011
10101001
10101001
10101001
101011001111011

9- 10101011
10001101
10101011
10101011
10101011
10101011
101111110101111

10- 11111111
10100001
11111111
11111111
11111111
101111111011111

DIVISIÓN

a- 100100 | 111
111 — 0101
001000
111
0001

b- 111001 | 101
101 — 1011
01000
101
00111
101
10

c- 1101 | 1010
1010 — 1
0011

d- 100000 | 100
100 — 1000
0000

e- 101101 | 111
111 — 0110
1000
111
0011

F- 101101 | 101
101 — 1001
0101
101
0

G- 111111 | 101
101 — 1100
101
101
011

H- 100001 | 101
101 — 0111
0110
101
101
101
0

i- 101101 | 110
110 — 0111
1010
110
1001
110
011

J- 100101 | 100
100 — 1001
0101
100
1

OCTAL

SUMA

1- 3712 + 144 = 4056
2- 444 + 35 = 501
3- 777 + 101 = 1100
4- 1025 + 777 = 2024
5- 525 + 252 = 777
6- 467 + 67 = 556
7- 2652 + 2652 = 5524
8- 1177 + 7711 = 11110
9- 1515 + 5151 = 6666
10- 1212 + 2121 = 3333

RESTA

1- 3712 − 144 = 3546
2- 444 − 35 = 407
3- 333 − 25 = 306
4- 777 − 101 = 676
5- 1025 − 777 = 0026
6- 525 − 252 = 253
7- 467 − 67 = 400
8- 2652 − 1001 = 1651
9- 1177 − 177 = 1000
10- 1515 − 151 = 1344

MULTIPLICACIÓN

1- 45 × 5 = 271
2- 357 × 10 = 3570
3- 162 × 15
4- 107 × 25
5- 3712 × 144
17450
17450
3712
1147350
6- 444 × 35
2664
1554
20424
7- 1200 × 300 = 360000
8- 1515 × 152
3232
10201
1515
256742
9- 1212 × 121
1212
1414
1212
136552
10- 777 × 111
777
777
777
110667

DIVISION

1- 37 ⌊4 → 34, 7, 3/
2- 10 ⌊5 → 5, 1, 3/
3- 455 ⌊150 → 320, 2, 135/
4- 600 ⌊77 → 572, 6, 6/
5- 1212 ⌊121 → 121, 10, 02/
6- 1515 ⌊151 → 151, 10, 05/
7- 777 ⌊77 → 77, 10, 07/
8- 1532 ⌊153 → 153, 10, 02/
9- 1001 ⌊110 → 770, 7, 11/
10- 1111 ⌊111 → 111, 10, 01/

HEXADECIMAL SUMA

1- AFF9 + 25C = B255
2- FF + 75E = 85D
3- 499F + 27A = 4C99
4- 9FC + 10A = B06
5- 77FF + 12CD = 8ACC
6- 152A + BB = 15E5
7- 1212 + FF = 1311
8- 1515 + CDEE = E303
9- 100B + 100C = 2017
10- 100 + 101 = 201

RESTA

1- AFF9 − 25C = AD9D
2- FF − 75 = 8A
3- 499F − 27A = 4725
4- 9FC − 10A = 8F2
5- 77FF − 12C = 76D3
6- 152A − BB = 146F
7- 1212 − FF = 1113
8- 100B − 100 = F0B
9- 9AA9 − 1212 = 8897
10- 10001 − 1110 = EEF1

MULTIPLICACIÓN

1- 1212 × A12
2424
1212
B4B4
B5F944

2- 1212 × BB
C6C6
C6C6
D3326

3- 100BBB × EFE
E0A43A
F0AFF5
E0A43A
F0BFDD8A

4- BBB100 × 15A
754EA00
3AA7500
BBB100
FDAD3A00

5- FF77 × 12
FEEE
FF77
10F65E

6- 10110 × AE
E0EE0
A0AA0
AEBBE0

7- 10001 × 12B
B000B
20002
10001
12B012B

8- 11110 × 397A
AAAA0
77770
99990
33330
3D4EB1A0

9- 1959E

$$\begin{array}{r} \times \quad FF \\ \hline 17C442 \\ 17C442 \\ \hline 1940862 \end{array}$$

10- 1515

$$\begin{array}{r} \times \quad EF \\ \hline 13C3B \\ 12726 \\ \hline 13AE9B \end{array}$$

DIVISIÓN

1- 101A ∟101
101 10
0A/

2- ABC ∟15
A8 82
3C
2A
12/

3- BCD ∟16A
B50 8
07D

4- CDE ∟178
BD8 8
106/

5- DEF ∟18C
DEC 9
3/

6- FF ∟CC
CC 1
33/

7- FF9 ∟100
F00 F
F9/

8- 89E ∟10A
850 8
4E

9- 1234 ∟123
1230 16
4/

10- 457 ∟A
3C 6F
97
96
1/

A- Decimal a binario

1243 617 308 154 77 38 19 9 4 2 1 = 10011010010

0 1 0 0 1 0 1 1 0 0 1

2345 1172 586 293 146 73 36 18 9 4 2 1 = 100100101001

1 0 0 1 0 1 0 0 1 0 0 1

3456 1728 864 432 216 108 54 27 13 6 3 1 = 110110000000

0 0 0 0 0 0 0 1 1 0 1 1

4567 2283 1141 570 285 142 71 35 17 8 4 2 1 = 1000111010111

1 1 1 0 1 0 1 1 1 0 0 0 1

5678 2839 1419 709 354 177 88 44 22 11 5 2 1 = 1011000101110

0 1 1 1 0 1 0 0 0 1 1 0 1

B- Decimal a octal

1234 |8 = 2322
2/ 154 |8
2/ 19 |8
3/ 2

2345 |8 = 4451
1/ 293 |8
5/ 36 |8
4/ 4

3456 |8 = 6600
0/ 432 |8
0/ 54 |8
6/ 6

4567 |8 = 10727
7/ 570 |8
2/ 71 |8
7/ 8 |8
0 1

5678 |8 = 13056
6/ 709 |8
5/ 88 |8
0/ 11 |8
3/ 1

C- Decimal a hexadecimal

1234 |16 = 4D2
2/ 77 |16
D/ 4

2345 |16 = 929
9/ 146 |16
2/ 9

3456 |16 = D80
0/ 216 |16
8/ D

4567 |16 = 11D7
7/ 285 |16
D/ 17 |16
1/ 1

5678 |16 = 162E
E/ 354 |16
2/ 22 |16
6/ 1

D- Binario a decimal

10101010 = 170
128 64 32 16 8 4 2 1

11110000 = 240

11001100 = 204

11011101 = 221

E- Binario a Octal

010101010 → 2 5 2

011110000 → 3 6 0

011001100 → 3 1 4

011011101 → 3 3 5

011111111111111110000000001010101 → 3 7 7 7 7 6 0 0 1 2 5

F- Binario a Hexadecimal

$\underbrace{1010}_{A}\underbrace{1010}_{A}$ $\underbrace{1111}_{F}\underbrace{0000}_{0}$ $\underbrace{1100}_{C}\underbrace{1100}_{C}$ $\underbrace{1101}_{D}\underbrace{1101}_{D}$

$\underbrace{1111}_{F}\underbrace{1111}_{F}\underbrace{1111}_{F}\underbrace{1111}_{F}\underbrace{0000}_{0}\underbrace{0000}_{0}\underbrace{0101}_{5}\underbrace{0101}_{5}$

G-Octal a Decimal

7535= 3933 741212 = 2464012345 = 5349 14563 = 6515

2345= 1253 1425= 789 2536= 1374 3636= 1950

49231223 = 9777811 125637= 43935

H-Octal a binario

7535 = $\overbrace{111}^{7}\overbrace{101}^{5}\overbrace{011}^{3}\overbrace{101}^{5}$

741212= 111 100 001 020 001 010

12345= 001 010 011 100 101

14563= 001 100 101 110 011

2345= 010 011 100 101

1425= 001 100 010 101

2536= 010 101 011 110

3636= 011 110 011 110

45231223= 100 101 010 011 001 010 010 011

125637= 001 010 101 110 011 111

i-Octal a hexadecimal

7535 = 111101011101 = FAD

741212 = 001111000010 10001010 = 3C28A

12345 = 001010011100101 = 14E5

14563 = 0001100101110011 = 1973

2345 = 0010 0011 0100 0101 = 4E5

1425 = 0001 0100 0010 0101 = 315

2536 = 1010101111 0 = 55E

3636 = 11110011110 = 79E

45231223 = 1001010100110010 10010011 = 953293

125637 = 0010101011100 11111 = 0AB9F

J-Hexadecimal a decimal

1234= 4660

2345= 9029

12ABC= 76.476

45CDB= 285915

99FF= 39423

100EEA= 1052394

A12A= 41258

99EEF= 630511

7979= 31097

8181B= 530459

K-Hexadecimal a binario

1234= 1001000110100

2345= 10001101000101

12ABC 10010101010111100

45CDB= 1000101110011011011

99FF= 1001100111111111

100EEA= 100000000111011101010

A12A= 1010000100101010

99EEF= 1001 1001 1110 1110 1111

7979= 111100101111001

8181B= 10000001100000011011

L-Hexadecimal a octal

1234 = 1001000110100 = 11064

2345 = 11001101000101 = 31505

12ABC= 10010101010111100 = 225274

45CDB = 1000101 1100 1101 1011 = 1056333

99FF= 1001100111111111 = 114777

100EEA= 100000000111011101010= 4007352

A12A= 1010000100101010 = 120452

99EEF= 10011001111011101111= 2317357

7979= 0111100101111001= 074571

8181B= 10000001100000011011= 2014033

Complemento A2

1- 11001 = 00111

10001011 = 01110101

110011010 = 001100110

111010111 = 000101001

2- 11010001101

01000111101

111 1111

11010001101

\-

10111000011

$\boxed{1}$ 10001010000

10110011101

\- 00001110101

111 11111

10110011101

11110001011

$\boxed{1}$ 10100101000